

# GammaLoop

Differential cross-sections with Local Unitarity

## 1 Overview

GammaLoop computes differential collider cross-sections with the Local Unitarity method. It is an application and research codebase rather than a thin numerical library: process generation, model import, integrand construction, integration, observables, persistence, and diagnostics meet in one stateful workflow.

### Primary interface

The command-line interface is the primary user interface. A run card creates a state, executes commands, and leaves a reusable working directory. The Rust and Python APIs load the same persisted state and expose selected structured operations; they do not define a second, independent execution model.

### 1.1 A stateful run

A typical source checkout is built and exercised from the repository root:

```
just build-cli-release
./gammaLoop ./examples/cli/gg_hhh/1L/gg_hhh_1L.toml
```

The run card imports a model, generates the requested process, executes its command blocks, and records the resulting state. Resume that directory explicitly for subsequent work:

```
./gammaLoop -s ./examples/cli/gg_hhh/1L/state \
  run integrate_physical -c "quit -o"
```

The persisted run.toml records how to replay the run. The settings files describe global and default runtime configuration, while processes/ contains process-specific state. Ordinary runs create gammaLoop\_state/ unless a different state path is supplied.

### 1.2 Installation and external tools

GammaLoop is currently developed primarily from source. The supported development path is the repository's Nix shell, or a local Rust toolchain together with just, a recent GNU toolchain, and Python 3.11 or newer when building bindings. FORM 4.2.1 or newer is needed for analytical integration of integrated UV counterterms. UFO model import also needs the Python ufo-model-loader package.

```
nix develop
just build-cli
just build-api
```

Diagram rendering is a separate concern and uses Clinnet and Typst. Building the CLI does not imply that these drawing tools are installed.

## 1.3 Related crates

### How the crates fit together

Linnet provides the half-edge graph model and graph algorithms. Spenso provides typed tensors, tensor structures, and network execution. Idenso handles symbolic tensor identities and algebraic simplification. Vakint matches and evaluates vacuum-integral topologies. GammaLoop combines these components into a collider-calculation workflow.

For a deeper view of how the command layer, state, integrands, and external components fit together, see the [architecture overview](#).

## 1.4 Where to begin

- Use built-in `--help` for the CLI options supported by your installed version.
- Start with the included `gg_hhh` run card for a complete state lifecycle.
- Use the Rust or Python API only when a program needs structured access to loaded state or sample-evaluation results.
- Check prerequisites before starting an expensive integration; some workflows require a Symbolica license or external tools such as FORM.

## 2 Tutorial

This tutorial builds the GammaLoop command-line interface and uses the included one-loop `g g -> h h h` run card to create a state that can be inspected and resumed. Compilation and process generation can take longer than a `--help` smoke test.

### 2.1 Prerequisites

Work from a GammaLoop source checkout. The supported development environment is the repository Nix shell; without Nix you need Rust 1.85 or newer, just, a recent GNU toolchain, and the dependencies described in the root README. The bundled Standard Model used below does not need the UFO loader. A Symbolica license may be required by your build and use case.

```
nix develop
just build-cli
./gammaloop --help
```

The last command should print the one-shot CLI options and command tree. The `./gammaloop` wrapper selects the binary built under `target/dev-optimize/`.

### 2.2 Generate an example process

The run card declares a state folder, settings, and named command blocks. Run its `generate` block and persist the result on exit:

```
./gammaloop --clean-state \
  ./examples/cli/gg_hhh/1L/gg_hhh_1L.toml \
  run generate -c "quit -o"
```

#### `--clean-state` removes the resolved state first`

Use it only for this first, reproducible run. Omit it when resuming work you want to keep. The example card resolves its state to `examples/cli/gg_hhh/1L/state` and requests ten worker cores; make a private copy of the card before changing either value.

The block imports `sm-default.json`, generates the selected one-loop pentagon contribution, builds its integrand, saves DOT data, and writes the state. A successful run exits without an error and leaves at least `run.toml`, `global_settings.toml`, `default_runtime_settings.toml`, and `processes/` in the state folder. `run.toml` records the commands and settings needed to replay the run; it is not merely a log file.

## 2.3 Resume the saved state

Load the saved state and ask the active session to display its processes:

```
./gammaloop -s ./examples/cli/gg_hhh/1L/state \  
run -c "display processes; quit -o"
```

Success means the display includes the generated `gg_hhh` process and the command exits while keeping the same state. You can now run the card's `integrate_euclidean` or `integrate_physical` block, but those are deliberately beyond the first-success path because they request a substantial Monte Carlo integration.

## 2.4 Troubleshooting and next steps

- If the wrapper cannot find a binary, rerun `just build-cli` and confirm that `target/dev-optim/gammaloop` exists.
- If compilation or process generation exhausts local resources, copy the card and reduce the values under `cli_settings.global.n_cores`; do not edit a state midway through generation.
- If startup reports a state fingerprint or settings mismatch, replay the card with a new state folder instead of transplanting only its `processes/` directory.
- Use `./gammaloop help generate` to inspect the flags supported by your installed version, then continue with the process-generation manual and the CLI reference.

### Example run card

See the complete `gg_hhh` run card before adapting the process, state location, or resource settings for your own calculation.

# 3 Process generation and state workflows

GammaLoop describes a calculation with a TOML run card and stores its reusable state in a directory. Start the command-line interface through the repository wrapper (`./gammaloop`) or the Cargo-built executable, then generate processes, integrands, and integration jobs within that state.

## 3.1 Choose a generation mode

The interactive command tree separates cross sections, amplitudes, and operations on processes already stored in the active state:

```
generate xs INITIAL > FINAL ...  
generate amp INITIAL > FINAL ...  
generate existing ...  
generate help xs
```

`xs` constructs forward-scattering graphs and filters compatible Cutkosky cuts. `amp` constructs ordinary amplitudes. Their final-state lists therefore have different meanings even when the printed particle syntax is identical. Use `generate help <mode>` inside the stateful CLI to see the flags supported by your installed version.

### 3.2 Process specification grammar

Process specifications accept `to` or `>` between space-separated initial and final states. Braces express alternatives in a cross-section final state, while `{}` denotes an empty state:

```
e+ e- > mu+ mu-
e+ e- > { Z Z, a a, H H }
{} > {}
```

A slash vetoes particles and a vertical bar selects an allow-list. The words `veto` and `only` are equivalent forms. Amplitude coupling constraints use the coupling name directly; powered constraints apply to the complete cross section:

```
e+ e- > d d~ g / u c QED==2 QCD>=2 QCD<=4
e+ e- > mu+ mu- | g ghG QED^2==2 QCD^2>=2
```

The bracketed perturbative block is intentionally distinct from ordinary flags. `{n}` fixes the amplitude loop count (or the sum across the two cut sides), `{{n}}` fixes the forward-graph loop count, and `QCD=n` or the shorthand `QCD` selects a relative perturbative order:

```
e+ e- > Z [ {1} {{2}} QCD=2 QED=1 ]
e+ e- > Z [ QCD ]
```

#### Resolve names through the active model

Particle names, antiparticles, coupling-order names, and allowed interactions come from the model loaded in the current state. A syntactically valid specification can still be rejected when that model does not define a requested name.

### 3.3 Filters, grouping, and multiplicity

Generation flags control topology filters, loop ranges, cut blobs and spectators, symmetrization, and numerator-aware grouping. These choices are physics-significant. In particular, grouping can combine graph multiplicities and numerator ratios; external-fermion symmetrization requires the caller to retain the associated ordering signs. Run `generate help <mode>` before choosing filters to see the available aliases, defaults, values, and argument counts.

The generated graph's overall factor keeps separate contributions for automorphisms, internal fermion loops, external-fermion ordering, numerator-independent symmetry grouping, and numerator-dependent grouping. Persist the state and inspect that factor when auditing diagram normalization instead of replacing it with an assumed factorial.

### 3.4 From generation to evaluation

A reproducible run keeps its stages explicit:

- `import` or select the model before resolving a process specification;
- `generate` the process and inspect the retained diagrams;
- `generate` the requested integrand representation;
- set kinematics, sampling, stability, and subtraction settings;
- evaluate samples or run an integration command block;
- exit with persistence enabled when the state should be resumed.

#### Keep the complete state together

`run.toml` records boot settings, reusable command blocks, and top-level commands. Generated process and

integrand artifacts live beside it. Resume with `-s STATE_FOLDER`; replay the run card to reconstruct the state from scratch. Copy or archive the entire state directory rather than moving a generated subdirectory on its own, because settings and fingerprints are checked together.

## 4 CLI, Rust, and Python APIs

### 4.1 Rust packages

The Rust API spans two primary packages.

- `gammalooprs` contains the physics implementation, data model, integrands, integration, observables, and lower-level algorithms.
- `gammaloop-api` provides the supported state-loading API, session/command integration, CLI assembly, and the PyO3 extension used by the Python distribution.

Load a state with `StateLoadOption::load()`. Its options select a state directory, boot card, logging overrides, read-only behavior, and optional settings overrides. Once loaded, callers can create a CLI-style session or use dedicated structured operations. A raw command string is the compatibility fallback, not a replacement for a typed operation where one exists.

```
use std::path::PathBuf;
use gammaloop_api::{state::CommandHistory, StateLoadOption};

let mut loaded = StateLoadOption {
    state_folder: Some(PathBuf::from("./state")),
    ..StateLoadOption::default()
}.load()?;

let command = CommandHistory::from_raw_string("display settings global"?);
loaded.cli_session().execute_command(command)?;
```

Some Rust types are public so that GammaLoop's packages can work together, but are not intended as stable integration points. Prefer the state-loading and structured-operation APIs described here; start with the [curated Rust API](#) and use its [Rustdoc links](#) when you need lower-level types or trait implementations.

### 4.2 Python packaging

#### A standalone GammaLoop distribution

The Python distribution and import package are both named `gammaloop`. Its compiled backend is `gammaloop._gammaloop`, which applications normally access through the public package. This is separate from the `symbolica.community.*` modules used by `Spenso`, `Idenso`, and `Vakint`; installing GammaLoop does not install those community modules as independent packages.

The main Python entry point is `GammaLoopAPI`:

```
from gammaloop import GammaLoopAPI

gl = GammaLoopAPI(
    state_folder="./state",
    boot_commands_path="./run.toml",
)
gl.run("display settings global")
```

The Python package requires Python 3.11 or newer. Run `just build-api` when building the bindings from a source checkout.

### 4.3 Sample evaluation and precision

`evaluate_sample` returns one sample result and the observable bundle for that one-sample batch. `evaluate_samples` returns per-sample results plus a batch-global observable bundle. Both the ordinary Rust and Python endpoints expose numeric results through an `f64` contract, even if arbitrary precision was used internally. Rust-only callers that must retain the active precision use `evaluate_sample_precise` and `evaluate_samples_precise`.

### 4.4 CLI and settings reference

Use `./gamma-loop --help` and the generated CLI and settings reference for command names, aliases, flags, positional arguments, defaults, possible values, and setting paths. The tutorials explain how commands and settings combine into a persistent workflow.

## 5 Releases and change history

GammaLoop versions are identified by repository tags together with the version of the Python distribution and the `gamma-loop-rs` Rust package. These components can evolve independently, so record all of them when reporting a result or asking for support.

### 5.1 Choosing the matching documentation

The `latest` channel follows current development. Documentation under `snapshots/<tag>` stays with a tagged release. Use the snapshot matching your checkout whenever command options, state files, or API signatures differ from `latest`.

### 5.2 Before upgrading

Release notes currently live with repository tags and their linked pull requests rather than in one consolidated changelog. Review them before upgrading, especially when you depend on persisted states, CLI scripts, Python bindings, or numerical results. Keep a copy of valuable state directories and re-run validation samples after changing versions.

#### Information to include in a report

Include the repository tag or commit, the Python distribution version, the `gamma-loop-rs` version, the run card, and any relevant settings overrides. This makes a calculation easier to reproduce and distinguishes a software change from a configuration change.

## API reference

The packages available in this version are listed below. Follow an item link for its complete language reference.

### gammaloops

GammaLoop's supported Rust entry points. The complete compiled surface remains available in the Rustdoc sidecar.

#### GammaLoopContext

Combines the state-map and model capabilities required by GammaLoop computations.

Combines the state-map and model capabilities required by GammaLoop computations.

```
pub trait GammaLoopContext : HasStateMap + HasModel {}
```

#### Supported usage

```
fn accepts_context<C: gammaloops::GammaLoopContext>(_context: &C) {}
```

Exhaustive reference · Source

#### HasModel

Provides shared access to the physics model owned by a computation context.

Provides shared access to the physics model owned by a computation context.

```
pub trait HasModel { fn get_model(& self) -> & Model; }
```

#### Supported usage

```
fn model_owner<C: gammaloops::HasModel>(_context: &C) {}
```

#### get\_model

**Kind:** Method

```
fn get_model(& self) -> & Model;
```

Exhaustive reference · Source

#### Install interrupt handling

Installs GammaLoop's process interrupt handler before a long-running integration.

Installs GammaLoop's process interrupt handler before a long-running integration.

```
fn set_interrupt_handler()
```

#### Supported usage

```
gammaloops::set_interrupt_handler();
```

Exhaustive reference · Source

### gammaloop-api

GammaLoop's stable facade and command API. Use this curated boundary for applications while consulting the full Rustdoc for exhaustive details.

#### State loading options

Selects the state folder, boot commands, settings overrides, logging, and read-only mode.

Selects the state folder, boot commands, settings overrides, logging, and read-only mode.

```
#[derive(Debug, Clone, PartialEq, Default)] pub struct StateLoadOption
{
    pub clean_state : bool, pub boot_commands_path : Option < PathBuf > , pub
    state_folder : Option < PathBuf > , pub model_file : Option < PathBuf > ,
    pub trace_logs_filename : Option < String > , pub level : Option <
    LogLevel > , pub logfile_level : Option < LogLevel > , pub logging_prefix
    : Option < LogFormat > , pub read_only_state : bool, pub
    settings_global_path : Option < PathBuf > , pub
    settings_runtime_defaults_path : Option < PathBuf > ,
}
```

**Reference feature matrix:** cli (item is available without an additional gate)

### Supported usage

```
let options = gammaloop_api::StateLoadOption::default();
```

#### **clean\_state**

**Kind:** Field

bool

#### **boot\_commands\_path**

**Kind:** Field

Option < PathBuf >

#### **state\_folder**

**Kind:** Field

Option < PathBuf >

#### **model\_file**

**Kind:** Field

Option < PathBuf >

#### **trace\_logs\_filename**

**Kind:** Field

Option < String >

#### **level**

**Kind:** Field

Option < LogLevel >

#### **logfile\_level**

**Kind:** Field

Option < LogLevel >

#### **logging\_prefix**

**Kind:** Field

Option < LogFormat >

#### **read\_only\_state**

**Kind:** Field

bool

## settings\_global\_path

**Kind:** Field

Option < PathBuf >

## settings\_runtime\_defaults\_path

**Kind:** Field

Option < PathBuf >

Exhaustive reference · Source

## Load a GammaLoop state

Loads the selected state and returns the supported session boundary.

Loads the selected state and returns the supported session boundary.

```
fn load(self) -> Result < LoadedState >
```

**Reference feature matrix:** cli (item is available without an additional gate)

| Parameter | Type |
|-----------|------|
| self      | self |

**Returns:** Result < LoadedState >

## Supported usage

```
let loaded = gammaloop_api::StateLoadOption::default().load()?;
```

Exhaustive reference · Source

## Create a CLI session

Borrows a loaded state as a command session while keeping state ownership explicit.

Borrows a loaded state as a command session while keeping state ownership explicit.

```
fn cli_session(& mut self) -> CliSession < '_ >
```

**Reference feature matrix:** cli (item is available without an additional gate)

| Parameter | Type       |
|-----------|------------|
| self      | & mut self |

**Returns:** CliSession < '\_ >

## Supported usage

```
let mut loaded = gammaloop_api::StateLoadOption::default().load()?; let session =  
loaded.cli_session();
```

Exhaustive reference · Source

## Command-line settings

Owns the generated command-line, global, session, and state-loading settings.

Owns the generated command-line, global, session, and state-loading settings.

```
#[derive(Debug, Clone, Deserialize, Serialize, JsonSchema)]  
#[serde(default, deny_unknown_fields)] pub struct CLISettings  
{  
    #[serde(skip_serializing_if = "is_true")] pub try_strings : bool,  
    #[serde(skip_serializing_if = "is_false")] pub override_state : bool,
```

```

    #[serde(skip_serializing_if = "IsDefault::is_default")] pub state :
    StateSettings, #[serde(skip_serializing_if = "IsDefault::is_default")] pub
    global : GlobalSettings, #[serde(skip)] #[schemars(skip)] pub session :
    SessionSettings,
}

```

**Reference feature matrix:** cli (item is available without an additional gate)

### Supported usage

```
let settings = gammaloop_api::CLISettings::default();
```

#### try\_strings

**Kind:** Field

bool

#### override\_state

**Kind:** Field

bool

#### state

**Kind:** Field

StateSettings

#### global

**Kind:** Field

GlobalSettings

#### session

**Kind:** Field

SessionSettings

Exhaustive reference · Source

### GammaLoop command

Runs the Clap-derived command tree for state creation, inspection, generation, and evaluation.

Runs the Clap-derived command tree for state creation, inspection, generation, and evaluation.

```
gammaloop [OPTIONS] [COMMAND]
```

**Reference feature matrix:** cli (item is available without an additional gate)

### Supported usage

```
gammaloop --help
```

#### try\_strings

**Kind:** Field

bool

#### override\_state

**Kind:** Field

bool

## **state**

**Kind:** Field

StateSettings

## **global**

**Kind:** Field

GlobalSettings

## **session**

**Kind:** Field

SessionSettings

Exhaustive reference · Source

## **gammaloop-python**

Exports registered by gammaloop.

### **GammaLoopAPI**

Stateful entry point for loading, evaluating, and integrating GammaLoop processes.

Stateful entry point for loading, evaluating, and integrating GammaLoop processes.

class GammaLoopAPI:

**Requires features:** python\_api, python\_abi, python\_stubgen

### **Inspect the registered export**

```
from gammaloop._gammaloop import GammaLoopAPI
help(GammaLoopAPI)
```

### **\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, state_folder: typing.Optional[builtins.str | os.PathLike |
pathlib.Path] = None, boot_commands_path: typing.Optional[builtins.str | os.PathLike
| pathlib.Path] = None, model_file: typing.Optional[builtins.str | os.PathLike |
pathlib.Path] = None, trace_logs_filename: typing.Optional[builtins.str] = None,
level: typing.Optional[LogLevel] = None, logfile_level: typing.Optional[LogLevel] =
None, logging_prefix: builtins.object | None = None, read_only_state: builtins.bool
= ..., settings_global_path: typing.Optional[builtins.str | os.PathLike |
pathlib.Path] = None, settings_runtime_defaults_path: typing.Optional[builtins.str |
os.PathLike | pathlib.Path] = None, clean_state: builtins.bool = ...) ->
GammaLoopAPI:
```

### **state\_folder**

**Kind:** Parameter

state\_folder: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None

**Default:** None

### **boot\_commands\_path**

**Kind:** Parameter

boot\_commands\_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None

**Default:** None

**model\_file****Kind:** Parameter`model_file: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None`**Default:** None**trace\_logs\_filename****Kind:** Parameter`trace_logs_filename: typing.Optional[builtins.str] = None`**Default:** None**level****Kind:** Parameter`level: typing.Optional[LogLevel] = None`**Default:** None**logfile\_level****Kind:** Parameter`logfile_level: typing.Optional[LogLevel] = None`**Default:** None**logging\_prefix****Kind:** Parameter`logging_prefix: builtins.object | None = None`**Default:** None**read\_only\_state****Kind:** Parameter`read_only_state: builtins.bool = ...`**Default:** ...**settings\_global\_path****Kind:** Parameter`settings_global_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None`**Default:** None**settings\_runtime\_defaults\_path****Kind:** Parameter`settings_runtime_defaults_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None`**Default:** None**clean\_state****Kind:** Parameter`clean_state: builtins.bool = ...`

**Default:** ...

### **evaluate\_sample**

**Kind:** Method

```
def evaluate_sample(self, point: typing.Sequence[builtins.float], process_id:
typing.Optional[builtins.int] = None, integrand_name: typing.Optional[builtins.str] =
None, use_arb_prec: builtins.bool = ..., minimal_output: builtins.bool = ...,
return_events: typing.Optional[builtins.bool] = None, momentum_space: builtins.bool
= ..., integrator_weight: typing.Optional[builtins.float] = None, discrete_dim:
typing.Optional[typing.Sequence[builtins.int]] = None, graph_name:
typing.Optional[builtins.str] = None, orientation: typing.Optional[builtins.int] =
None) -> EvaluationResult:
```

#### **point**

**Kind:** Parameter

point: typing.Sequence[builtins.float]

#### **process\_id**

**Kind:** Parameter

process\_id: typing.Optional[builtins.int] = None

**Default:** None

#### **integrand\_name**

**Kind:** Parameter

integrand\_name: typing.Optional[builtins.str] = None

**Default:** None

#### **use\_arb\_prec**

**Kind:** Parameter

use\_arb\_prec: builtins.bool = ...

**Default:** ...

#### **minimal\_output**

**Kind:** Parameter

minimal\_output: builtins.bool = ...

**Default:** ...

#### **return\_events**

**Kind:** Parameter

return\_events: typing.Optional[builtins.bool] = None

**Default:** None

#### **momentum\_space**

**Kind:** Parameter

momentum\_space: builtins.bool = ...

**Default:** ...

**integrator\_weight****Kind:** Parameter

integrator\_weight: typing.Optional[builtins.float] = None

**Default:** None**discrete\_dim****Kind:** Parameter

discrete\_dim: typing.Optional[typing.Sequence[builtins.int]] = None

**Default:** None**graph\_name****Kind:** Parameter

graph\_name: typing.Optional[builtins.str] = None

**Default:** None**orientation****Kind:** Parameter

orientation: typing.Optional[builtins.int] = None

**Default:** None**evaluate\_samples****Kind:** Method

```
def evaluate_samples(self, points: numpy.typing.NDArray[numpy.float64], process_id:
typing.Optional[builtins.int] = None, integrand_name: typing.Optional[builtins.str] =
None, use_arb_prec: builtins.bool = ..., minimal_output: builtins.bool = ...,
return_events: typing.Optional[builtins.bool] = None, momentum_space: builtins.bool
= ..., integrator_weights: typing.Optional[numpy.typing.NDArray[numpy.float64]] =
None, discrete_dims: numpy.typing.NDArray[numpy.unsignedinteger] | None = None,
graph_names: typing.Optional[typing.Sequence[typing.Optional[builtins.str]]] = None,
orientations: typing.Optional[typing.Sequence[typing.Optional[builtins.int]]] = None)
-> BatchEvaluationResult:
```

**points****Kind:** Parameter

points: numpy.typing.NDArray[numpy.float64]

**process\_id****Kind:** Parameter

process\_id: typing.Optional[builtins.int] = None

**Default:** None**integrand\_name****Kind:** Parameter

integrand\_name: typing.Optional[builtins.str] = None

**Default:** None

**use\_arb\_prec****Kind:** Parameter

use\_arb\_prec: builtins.bool = ...

**Default:** ...**minimal\_output****Kind:** Parameter

minimal\_output: builtins.bool = ...

**Default:** ...**return\_events****Kind:** Parameter

return\_events: typing.Optional[builtins.bool] = None

**Default:** None**momentum\_space****Kind:** Parameter

momentum\_space: builtins.bool = ...

**Default:** ...**integrator\_weights****Kind:** Parameter

integrator\_weights: typing.Optional[numpy.typing.NDArray[numpy.float64]] = None

**Default:** None**discrete\_dims****Kind:** Parameter

discrete\_dims: numpy.typing.NDArray[numpy.unsignedinteger] | None = None

**Default:** None**graph\_names****Kind:** Parameter

graph\_names: typing.Optional[typing.Sequence[typing.Optional[builtins.str]]] = None

**Default:** None**orientations****Kind:** Parameter

orientations: typing.Optional[typing.Sequence[typing.Optional[builtins.int]]] = None

**Default:** None**import\_graphs****Kind:** Method

```
def import_graphs(self, graphs: builtins.str, process_name:
typing.Optional[builtins.str] = None, process_id: typing.Optional[builtins.int] =
```

```
None, integrand_name: typing.Optional[builtins.str] = None, format: builtins.str
= ..., overwrite: builtins.bool = ..., append: builtins.bool = ...) -> None:
```

### **graphs**

**Kind:** Parameter

graphs: builtins.str

### **process\_name**

**Kind:** Parameter

process\_name: typing.Optional[builtins.str] = None

**Default:** None

### **process\_id**

**Kind:** Parameter

process\_id: typing.Optional[builtins.int] = None

**Default:** None

### **integrand\_name**

**Kind:** Parameter

integrand\_name: typing.Optional[builtins.str] = None

**Default:** None

### **format**

**Kind:** Parameter

format: builtins.str = ...

**Default:** ...

### **overwrite**

**Kind:** Parameter

overwrite: builtins.bool = ...

**Default:** ...

### **append**

**Kind:** Parameter

append: builtins.bool = ...

**Default:** ...

### **get\_lmbs**

**Kind:** Method

```
def get_lmbs(self, graphs: builtins.str, format: builtins.str = ...) ->
builtins.list[builtins.list[tuple[builtins.list[builtins.int],
builtins.list[builtins.int], builtins.dict[builtins.int,
tuple[builtins.list[builtins.int], builtins.list[builtins.int]]]]]]:
```

### **graphs**

**Kind:** Parameter

graphs: builtins.str

**format****Kind:** Parameter

format: builtins.str = ...

**Default:** ...**get\_orientations****Kind:** Method

```
def get_orientations(self, graph_name: builtins.str, process_id:
typing.Optional[builtins.int] = None, integrand_name: typing.Optional[builtins.str] =
None) -> builtins.list[builtins.dict[builtins.int, builtins.int]]:
```

**graph\_name****Kind:** Parameter

graph\_name: builtins.str

**process\_id****Kind:** Parameter

process\_id: typing.Optional[builtins.int] = None

**Default:** None**integrand\_name****Kind:** Parameter

integrand\_name: typing.Optional[builtins.str] = None

**Default:** None**get\_model****Kind:** Method

```
def get_model(self) -> builtins.str:
```

**evaluate****Kind:** Method

```
def evaluate(self, process_id: typing.Optional[builtins.int] = None,
graphs_group_name: typing.Optional[builtins.str] = None, result_path:
typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None, numerical:
builtins.bool = ..., number_of_terms_in_epsilon_expansion:
typing.Optional[builtins.int] = None) -> builtins.str:
```

**process\_id****Kind:** Parameter

process\_id: typing.Optional[builtins.int] = None

**Default:** None**graphs\_group\_name****Kind:** Parameter

graphs\_group\_name: typing.Optional[builtins.str] = None

**Default:** None

**result\_path****Kind:** Parameter`result_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None`**Default:** None**numerical****Kind:** Parameter`numerical: builtins.bool = ...`**Default:** ...**number\_of\_terms\_in\_epsilon\_expansion****Kind:** Parameter`number_of_terms_in_epsilon_expansion: typing.Optional[builtins.int] = None`**Default:** None**import\_model****Kind:** Method`def import_model(self, model_specifier: builtins.str | os.PathLike | pathlib.Path, simplify_model: builtins.bool = ...) -> None:`**model\_specifier****Kind:** Parameter`model_specifier: builtins.str | os.PathLike | pathlib.Path`**simplify\_model****Kind:** Parameter`simplify_model: builtins.bool = ...`**Default:** ...**list\_outputs****Kind:** Method`def list_outputs(self) -> tuple[builtins.dict[builtins.str, builtins.int], builtins.dict[builtins.str, builtins.int]]:`**get\_integrand\_info****Kind:** Method`def get_integrand_info(self, process_id: typing.Optional[builtins.int] = None, integrand_name: typing.Optional[builtins.str] = None) -> IntegrandInfo:`**process\_id****Kind:** Parameter`process_id: typing.Optional[builtins.int] = None`**Default:** None**integrand\_name****Kind:** Parameter

integrand\_name: typing.Optional[builtins.str] = None

**Default:** None

#### **get\_integrand\_settings**

**Kind:** Method

```
def get_integrand_settings(self, process_id: typing.Optional[builtins.int] = None,
integrand_name: typing.Optional[builtins.str] = None) -> SettingsValue:
```

#### **process\_id**

**Kind:** Parameter

process\_id: typing.Optional[builtins.int] = None

**Default:** None

#### **integrand\_name**

**Kind:** Parameter

integrand\_name: typing.Optional[builtins.str] = None

**Default:** None

#### **get\_run\_history**

**Kind:** Method

```
def get_run_history(self) -> builtins.str:
```

#### **get\_global\_settings**

**Kind:** Method

```
def get_global_settings(self) -> builtins.str:
```

#### **get\_active\_command\_blocks**

**Kind:** Method

```
def get_active_command_blocks(self) -> builtins.dict[builtins.str,
builtins.list[builtins.str]]:
```

#### **get\_default\_runtime\_settings**

**Kind:** Method

```
def get_default_runtime_settings(self) -> SettingsValue:
```

#### **get\_dot\_files**

**Kind:** Method

```
def get_dot_files(self, process: typing.Optional[builtins.int | builtins.str] = None,
integrand_name: typing.Optional[builtins.str] = None, settings: DotExportSettings
= ...) -> builtins.str:
```

#### **process**

**Kind:** Parameter

process: typing.Optional[builtins.int | builtins.str] = None

**Default:** None

#### **integrand\_name**

**Kind:** Parameter

integrand\_name: typing.Optional[builtins.str] = None

**Default:** None

#### **settings**

**Kind:** Parameter

settings: DotExportSettings = ...

**Default:** ...

#### **run**

**Kind:** Method

def run(self, command: builtins.str) -> None:

#### **command**

**Kind:** Parameter

command: builtins.str

#### **generate\_cff**

**Kind:** Method

def generate\_cff(self, dot\_string: builtins.str, subgraph\_nodes: typing.Sequence[builtins.str], reverse\_dangling: typing.Sequence[builtins.int], orientation\_pattern: typing.Optional[builtins.str] = None) -> builtins.list[tuple[builtins.dict[builtins.int, builtins.int], builtins.str]]:

#### **dot\_string**

**Kind:** Parameter

dot\_string: builtins.str

#### **subgraph\_nodes**

**Kind:** Parameter

subgraph\_nodes: typing.Sequence[builtins.str]

#### **reverse\_dangling**

**Kind:** Parameter

reverse\_dangling: typing.Sequence[builtins.int]

#### **orientation\_pattern**

**Kind:** Parameter

orientation\_pattern: typing.Optional[builtins.str] = None

**Default:** None

#### **generate\_cff\_as\_json\_string**

**Kind:** Method

def generate\_cff\_as\_json\_string(self, dot\_string: builtins.str, subgraph\_nodes: typing.Sequence[builtins.str], reverse\_dangling: typing.Sequence[builtins.int], orientation\_pattern: typing.Optional[builtins.str] = None) -> builtins.str:

#### **dot\_string**

**Kind:** Parameter

dot\_string: builtins.str

#### **subgraph\_nodes**

**Kind:** Parameter

subgraph\_nodes: typing.Sequence[builtins.str]

#### **reverse\_dangling**

**Kind:** Parameter

reverse\_dangling: typing.Sequence[builtins.int]

#### **orientation\_pattern**

**Kind:** Parameter

orientation\_pattern: typing.Optional[builtins.str] = None

**Default:** None

[Exhaustive reference](#) · [Source](#)

#### **atom\_to\_canonical\_string**

Parse a Symbolica expression and return its canonical string form.

Parse a Symbolica expression and return its canonical string form.

```
def atom_to_canonical_string(atom_str: builtins.str) -> builtins.str:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

#### **Inspect the registered export**

```
from gammaloop._gammaloop import atom_to_canonical_string
help(atom_to_canonical_string)
```

[Exhaustive reference](#) · [Source](#)

#### **evaluate\_graph\_overall\_factor**

Evaluate a graph's symbolic overall factor and return its canonical form.

Evaluate a graph's symbolic overall factor and return its canonical form.

```
def evaluate_graph_overall_factor(overall_factor: builtins.str) -> builtins.str:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

#### **Inspect the registered export**

```
from gammaloop._gammaloop import evaluate_graph_overall_factor
help(evaluate_graph_overall_factor)
```

[Exhaustive reference](#) · [Source](#)

#### **to\_dots**

Rewrite a Symbolica tensor expression into Idenso dot-product notation.

Rewrite a Symbolica tensor expression into Idenso dot-product notation.

```
def to_dots(atom_str: builtins.str) -> builtins.str:
```

**Requires features:** python\_api, python\_abi, python\_stubgen

#### **Inspect the registered export**

```
from gammaloop._gammaloop import to_dots
help(to_dots)
```

Exhaustive reference · Source

## CLI commands and settings

Commands and settings available in this version are listed below.

### Command tree

| Command                                         | Description                                                                                         |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| gammaLoop                                       | An implementation of the Local Unitarity method for computing differential collider cross-sections. |
| gammaLoop display                               |                                                                                                     |
| gammaLoop display model                         |                                                                                                     |
| gammaLoop display processes                     |                                                                                                     |
| gammaLoop display integrand                     |                                                                                                     |
| gammaLoop display quantities                    |                                                                                                     |
| gammaLoop display observables                   |                                                                                                     |
| gammaLoop display selectors                     |                                                                                                     |
| gammaLoop display command_block                 |                                                                                                     |
| gammaLoop display settings                      |                                                                                                     |
| gammaLoop display settings global               |                                                                                                     |
| gammaLoop display settings default-runtime      |                                                                                                     |
| gammaLoop display settings process              |                                                                                                     |
| gammaLoop display settings help                 | Print this message or the help of the given subcommand(s)                                           |
| gammaLoop display settings help global          |                                                                                                     |
| gammaLoop display settings help default-runtime |                                                                                                     |
| gammaLoop display settings help process         |                                                                                                     |
| gammaLoop display settings help help            | Print this message or the help of the given subcommand(s)                                           |
| gammaLoop display help                          | Print this message or the help of the given subcommand(s)                                           |
| gammaLoop display help model                    |                                                                                                     |
| gammaLoop display help processes                |                                                                                                     |
| gammaLoop display help integrand                |                                                                                                     |
| gammaLoop display help quantities               |                                                                                                     |
| gammaLoop display help observables              |                                                                                                     |
| gammaLoop display help selectors                |                                                                                                     |
| gammaLoop display help command_block            |                                                                                                     |
| gammaLoop display help settings                 |                                                                                                     |

| Command                                         | Description                                                             |
|-------------------------------------------------|-------------------------------------------------------------------------|
| gammaLoop display help settings global          |                                                                         |
| gammaLoop display help settings default-runtime |                                                                         |
| gammaLoop display help settings process         |                                                                         |
| gammaLoop display help help                     | Print this message or the help of the given subcommand(s)               |
| gammaLoop duplicate                             |                                                                         |
| gammaLoop duplicate integrand                   |                                                                         |
| gammaLoop duplicate help                        | Print this message or the help of the given subcommand(s)               |
| gammaLoop duplicate help integrand              |                                                                         |
| gammaLoop duplicate help help                   | Print this message or the help of the given subcommand(s)               |
| gammaLoop set                                   |                                                                         |
| gammaLoop set base-dir                          | Set the base directory for gammaloop state and logs                     |
| gammaLoop set global                            | Set GLOBAL settings                                                     |
| gammaLoop set global file                       | Load from a settings file                                               |
| gammaLoop set global string                     | Load settings from TOML content passed as a CLI string                  |
| gammaLoop set global kv                         | Set one or more dotted key-paths                                        |
| gammaLoop set global defaults                   | Sync process runtime settings from the current default runtime settings |
| gammaLoop set global stored                     | Use the stored settings file                                            |
| gammaLoop set global help                       | Print this message or the help of the given subcommand(s)               |
| gammaLoop set global help file                  | Load from a settings file                                               |
| gammaLoop set global help string                | Load settings from TOML content passed as a CLI string                  |
| gammaLoop set global help kv                    | Set one or more dotted key-paths                                        |
| gammaLoop set global help defaults              | Sync process runtime settings from the current default runtime settings |
| gammaLoop set global help stored                | Use the stored settings file                                            |
| gammaLoop set global help help                  | Print this message or the help of the given subcommand(s)               |
| gammaLoop set default-runtime                   | Set DEFAULT RUNTIME settings                                            |
| gammaLoop set default-runtime file              | Load from a settings file                                               |
| gammaLoop set default-runtime string            | Load settings from TOML content passed as a CLI string                  |
| gammaLoop set default-runtime kv                | Set one or more dotted key-paths                                        |

| Command                                     | Description                                                             |
|---------------------------------------------|-------------------------------------------------------------------------|
| gammaLoop set default-runtime defaults      | Sync process runtime settings from the current default runtime settings |
| gammaLoop set default-runtime stored        | Use the stored settings file                                            |
| gammaLoop set default-runtime help          | Print this message or the help of the given subcommand(s)               |
| gammaLoop set default-runtime help file     | Load from a settings file                                               |
| gammaLoop set default-runtime help string   | Load settings from TOML content passed as a CLI string                  |
| gammaLoop set default-runtime help kv       | Set one or more dotted key-paths                                        |
| gammaLoop set default-runtime help defaults | Sync process runtime settings from the current default runtime settings |
| gammaLoop set default-runtime help stored   | Use the stored settings file                                            |
| gammaLoop set default-runtime help help     | Print this message or the help of the given subcommand(s)               |
| gammaLoop set model                         | Set Model parameters                                                    |
| gammaLoop set process                       | Set settings for a PROCESS                                              |
| gammaLoop set process file                  | Load from a settings file                                               |
| gammaLoop set process string                | Load settings from TOML content passed as a CLI string                  |
| gammaLoop set process kv                    | Set one or more dotted key-paths                                        |
| gammaLoop set process defaults              | Sync process runtime settings from the current default runtime settings |
| gammaLoop set process stored                | Use the stored settings file                                            |
| gammaLoop set process add                   | Add a named quantity/observable/selector                                |
| gammaLoop set process add quantity          |                                                                         |
| gammaLoop set process add observable        |                                                                         |
| gammaLoop set process add selector          |                                                                         |
| gammaLoop set process add help              | Print this message or the help of the given subcommand(s)               |
| gammaLoop set process add help quantity     |                                                                         |
| gammaLoop set process add help observable   |                                                                         |
| gammaLoop set process add help selector     |                                                                         |
| gammaLoop set process add help help         | Print this message or the help of the given subcommand(s)               |
| gammaLoop set process update                | Update a named quantity/observable/selector                             |

| Command                                      | Description                                                             |
|----------------------------------------------|-------------------------------------------------------------------------|
| gammaLoop set process update quantity        |                                                                         |
| gammaLoop set process update observable      |                                                                         |
| gammaLoop set process update selector        |                                                                         |
| gammaLoop set process update help            | Print this message or the help of the given subcommand(s)               |
| gammaLoop set process update help quantity   |                                                                         |
| gammaLoop set process update help observable |                                                                         |
| gammaLoop set process update help selector   |                                                                         |
| gammaLoop set process update help help       | Print this message or the help of the given subcommand(s)               |
| gammaLoop set process remove                 | Remove a named quantity/observable/selector                             |
| gammaLoop set process remove quantity        |                                                                         |
| gammaLoop set process remove observable      |                                                                         |
| gammaLoop set process remove selector        |                                                                         |
| gammaLoop set process remove help            | Print this message or the help of the given subcommand(s)               |
| gammaLoop set process remove help quantity   |                                                                         |
| gammaLoop set process remove help observable |                                                                         |
| gammaLoop set process remove help selector   |                                                                         |
| gammaLoop set process remove help help       | Print this message or the help of the given subcommand(s)               |
| gammaLoop set process help                   | Print this message or the help of the given subcommand(s)               |
| gammaLoop set process help file              | Load from a settings file                                               |
| gammaLoop set process help string            | Load settings from TOML content passed as a CLI string                  |
| gammaLoop set process help kv                | Set one or more dotted key-paths                                        |
| gammaLoop set process help defaults          | Sync process runtime settings from the current default runtime settings |
| gammaLoop set process help stored            | Use the stored settings file                                            |
| gammaLoop set process help add               | Add a named quantity/observable/selector                                |
| gammaLoop set process help add quantity      |                                                                         |

| Command                                      | Description                                                             |
|----------------------------------------------|-------------------------------------------------------------------------|
| gammaLoop set process help add observable    |                                                                         |
| gammaLoop set process help add selector      |                                                                         |
| gammaLoop set process help update            | Update a named quantity/observable/selector                             |
| gammaLoop set process help update quantity   |                                                                         |
| gammaLoop set process help update observable |                                                                         |
| gammaLoop set process help update selector   |                                                                         |
| gammaLoop set process help remove            | Remove a named quantity/observable/selector                             |
| gammaLoop set process help remove quantity   |                                                                         |
| gammaLoop set process help remove observable |                                                                         |
| gammaLoop set process help remove selector   |                                                                         |
| gammaLoop set process help help              | Print this message or the help of the given subcommand(s)               |
| gammaLoop set help                           | Print this message or the help of the given subcommand(s)               |
| gammaLoop set help base-dir                  | Set the base directory for gammaloop state and logs                     |
| gammaLoop set help global                    | Set GLOBAL settings                                                     |
| gammaLoop set help global file               | Load from a settings file                                               |
| gammaLoop set help global string             | Load settings from TOML content passed as a CLI string                  |
| gammaLoop set help global kv                 | Set one or more dotted key-paths                                        |
| gammaLoop set help global defaults           | Sync process runtime settings from the current default runtime settings |
| gammaLoop set help global stored             | Use the stored settings file                                            |
| gammaLoop set help default-runtime           | Set DEFAULT RUNTIME settings                                            |
| gammaLoop set help default-runtime file      | Load from a settings file                                               |
| gammaLoop set help default-runtime string    | Load settings from TOML content passed as a CLI string                  |
| gammaLoop set help default-runtime kv        | Set one or more dotted key-paths                                        |
| gammaLoop set help default-runtime defaults  | Sync process runtime settings from the current default runtime settings |
| gammaLoop set help default-runtime stored    | Use the stored settings file                                            |

| Command                                      | Description                                                             |
|----------------------------------------------|-------------------------------------------------------------------------|
| gammaLoop set help model                     | Set Model parameters                                                    |
| gammaLoop set help process                   | Set settings for a PROCESS                                              |
| gammaLoop set help process file              | Load from a settings file                                               |
| gammaLoop set help process string            | Load settings from TOML content passed as a CLI string                  |
| gammaLoop set help process kv                | Set one or more dotted key-paths                                        |
| gammaLoop set help process defaults          | Sync process runtime settings from the current default runtime settings |
| gammaLoop set help process stored            | Use the stored settings file                                            |
| gammaLoop set help process add               | Add a named quantity/observable/selector                                |
| gammaLoop set help process add quantity      |                                                                         |
| gammaLoop set help process add observable    |                                                                         |
| gammaLoop set help process add selector      |                                                                         |
| gammaLoop set help process update            | Update a named quantity/observable/selector                             |
| gammaLoop set help process update quantity   |                                                                         |
| gammaLoop set help process update observable |                                                                         |
| gammaLoop set help process update selector   |                                                                         |
| gammaLoop set help process remove            | Remove a named quantity/observable/selector                             |
| gammaLoop set help process remove quantity   |                                                                         |
| gammaLoop set help process remove observable |                                                                         |
| gammaLoop set help process remove selector   |                                                                         |
| gammaLoop set help help                      | Print this message or the help of the given subcommand(s)               |
| gammaLoop import                             |                                                                         |
| gammaLoop import model                       | Generate integrands                                                     |
| gammaLoop import graphs                      |                                                                         |
| gammaLoop import help                        | Print this message or the help of the given subcommand(s)               |
| gammaLoop import help model                  | Generate integrands                                                     |
| gammaLoop import help graphs                 |                                                                         |
| gammaLoop import help help                   | Print this message or the help of the given subcommand(s)               |
| gammaLoop save                               |                                                                         |

| Command                          | Description                                                 |
|----------------------------------|-------------------------------------------------------------|
| gammaLoop save dot               |                                                             |
| gammaLoop save uv-forest         |                                                             |
| gammaLoop save standalone        |                                                             |
| gammaLoop save state             |                                                             |
| gammaLoop save schema            | regenerate the schema files                                 |
| gammaLoop save help              | Print this message or the help of the given subcommand(s)   |
| gammaLoop save help dot          |                                                             |
| gammaLoop save help uv-forest    |                                                             |
| gammaLoop save help standalone   |                                                             |
| gammaLoop save help state        |                                                             |
| gammaLoop save help schema       | regenerate the schema files                                 |
| gammaLoop save help help         | Print this message or the help of the given subcommand(s)   |
| gammaLoop run                    |                                                             |
| gammaLoop start_commands_block   |                                                             |
| gammaLoop finish_commands_block  |                                                             |
| gammaLoop remove                 |                                                             |
| gammaLoop remove processes       |                                                             |
| gammaLoop remove help            | Print this message or the help of the given subcommand(s)   |
| gammaLoop remove help processes  |                                                             |
| gammaLoop remove help help       | Print this message or the help of the given subcommand(s)   |
| gammaLoop integrate              |                                                             |
| gammaLoop generate               | Generate integrands                                         |
| gammaLoop generate xs            | Cross-section (forward-scattering) generation               |
| gammaLoop generate amp           | Amplitude generation                                        |
| gammaLoop generate existing      | Reuse an already generated process                          |
| gammaLoop generate help          | Print this message or the help of the given subcommand(s)   |
| gammaLoop generate help xs       | Cross-section (forward-scattering) generation               |
| gammaLoop generate help amp      | Amplitude generation                                        |
| gammaLoop generate help existing | Reuse an already generated process                          |
| gammaLoop generate help help     | Print this message or the help of the given subcommand(s)   |
| gammaLoop select                 |                                                             |
| gammaLoop quit                   | Quit gammaloop                                              |
| gammaLoop inspect                | Inspect a single phase-space point / momentum configuration |
| gammaLoop approach               | Approach a phase-space point along one or more axes         |

| Command                                         | Description                                               |
|-------------------------------------------------|-----------------------------------------------------------|
| gammaLoop evaluate                              |                                                           |
| gammaLoop renormalize                           |                                                           |
| gammaLoop bench                                 | Benchmark raw integrand evaluation speed                  |
| gammaLoop profile                               |                                                           |
| gammaLoop profile ultra-violet                  | Ultraviolet profile analysis                              |
| gammaLoop profile bulk                          | Bulk profile analysis                                     |
| gammaLoop profile help                          | Print this message or the help of the given subcommand(s) |
| gammaLoop profile help ultra-violet             | Ultraviolet profile analysis                              |
| gammaLoop profile help bulk                     | Bulk profile analysis                                     |
| gammaLoop profile help help                     | Print this message or the help of the given subcommand(s) |
| gammaLoop batch                                 | HPC batch evaluation branch                               |
| gammaLoop !                                     |                                                           |
| gammaLoop help                                  | Print this message or the help of the given subcommand(s) |
| gammaLoop help display                          |                                                           |
| gammaLoop help display model                    |                                                           |
| gammaLoop help display processes                |                                                           |
| gammaLoop help display integrand                |                                                           |
| gammaLoop help display quantities               |                                                           |
| gammaLoop help display observables              |                                                           |
| gammaLoop help display selectors                |                                                           |
| gammaLoop help display command_block            |                                                           |
| gammaLoop help display settings                 |                                                           |
| gammaLoop help display settings global          |                                                           |
| gammaLoop help display settings default-runtime |                                                           |
| gammaLoop help display settings process         |                                                           |
| gammaLoop help duplicate                        |                                                           |
| gammaLoop help duplicate integrand              |                                                           |
| gammaLoop help set                              |                                                           |
| gammaLoop help set base-dir                     | Set the base directory for gammaloop state and logs       |
| gammaLoop help set global                       | Set GLOBAL settings                                       |
| gammaLoop help set global file                  | Load from a settings file                                 |

| Command                                      | Description                                                             |
|----------------------------------------------|-------------------------------------------------------------------------|
| gammaLoop help set global string             | Load settings from TOML content passed as a CLI string                  |
| gammaLoop help set global kv                 | Set one or more dotted key-paths                                        |
| gammaLoop help set global defaults           | Sync process runtime settings from the current default runtime settings |
| gammaLoop help set global stored             | Use the stored settings file                                            |
| gammaLoop help set default-runtime           | Set DEFAULT RUNTIME settings                                            |
| gammaLoop help set default-runtime file      | Load from a settings file                                               |
| gammaLoop help set default-runtime string    | Load settings from TOML content passed as a CLI string                  |
| gammaLoop help set default-runtime kv        | Set one or more dotted key-paths                                        |
| gammaLoop help set default-runtime defaults  | Sync process runtime settings from the current default runtime settings |
| gammaLoop help set default-runtime stored    | Use the stored settings file                                            |
| gammaLoop help set model                     | Set Model parameters                                                    |
| gammaLoop help set process                   | Set settings for a PROCESS                                              |
| gammaLoop help set process file              | Load from a settings file                                               |
| gammaLoop help set process string            | Load settings from TOML content passed as a CLI string                  |
| gammaLoop help set process kv                | Set one or more dotted key-paths                                        |
| gammaLoop help set process defaults          | Sync process runtime settings from the current default runtime settings |
| gammaLoop help set process stored            | Use the stored settings file                                            |
| gammaLoop help set process add               | Add a named quantity/observable/selector                                |
| gammaLoop help set process add quantity      |                                                                         |
| gammaLoop help set process add observable    |                                                                         |
| gammaLoop help set process add selector      |                                                                         |
| gammaLoop help set process update            | Update a named quantity/observable/selector                             |
| gammaLoop help set process update quantity   |                                                                         |
| gammaLoop help set process update observable |                                                                         |
| gammaLoop help set process update selector   |                                                                         |
| gammaLoop help set process remove            | Remove a named quantity/observable/selector                             |

| Command                                         | Description                                                 |
|-------------------------------------------------|-------------------------------------------------------------|
| gammaLoop help set process<br>remove quantity   |                                                             |
| gammaLoop help set process<br>remove observable |                                                             |
| gammaLoop help set process<br>remove selector   |                                                             |
| gammaLoop help import                           |                                                             |
| gammaLoop help import model                     | Generate integrands                                         |
| gammaLoop help import graphs                    |                                                             |
| gammaLoop help save                             |                                                             |
| gammaLoop help save dot                         |                                                             |
| gammaLoop help save uv-forest                   |                                                             |
| gammaLoop help save standalone                  |                                                             |
| gammaLoop help save state                       |                                                             |
| gammaLoop help save schema                      | regenerate the schema files                                 |
| gammaLoop help run                              |                                                             |
| gammaLoop help<br>start_commands_block          |                                                             |
| gammaLoop help<br>finish_commands_block         |                                                             |
| gammaLoop help remove                           |                                                             |
| gammaLoop help remove processes                 |                                                             |
| gammaLoop help integrate                        |                                                             |
| gammaLoop help generate                         | Generate integrands                                         |
| gammaLoop help generate xs                      | Cross-section (forward-scattering) generation               |
| gammaLoop help generate amp                     | Amplitude generation                                        |
| gammaLoop help generate existing                | Reuse an already generated process                          |
| gammaLoop help select                           |                                                             |
| gammaLoop help quit                             | Quit gammaloop                                              |
| gammaLoop help inspect                          | Inspect a single phase-space point / momentum configuration |
| gammaLoop help approach                         | Approach a phase-space point along one or more axes         |
| gammaLoop help evaluate                         |                                                             |
| gammaLoop help renormalize                      |                                                             |
| gammaLoop help bench                            | Benchmark raw integrand evaluation speed                    |
| gammaLoop help profile                          |                                                             |
| gammaLoop help profile ultra-violet             | Ultraviolet profile analysis                                |
| gammaLoop help profile bulk                     | Bulk profile analysis                                       |
| gammaLoop help batch                            | HPC batch evaluation branch                                 |
| gammaLoop help !                                |                                                             |

| Command             | Description                                               |
|---------------------|-----------------------------------------------------------|
| gammaLoop help help | Print this message or the help of the given subcommand(s) |

## Settings

| Path                                                                        | Type    | Default               |
|-----------------------------------------------------------------------------|---------|-----------------------|
| cli.global.display_directive                                                | string  | "info"                |
| cli.global.generation.compile.compilation_mode                              | string  | "symjit"              |
| cli.global.generation.compile.compiler                                      | string  | "g++"                 |
| cli.global.generation.compile.custom_array                                  | array   | []                    |
| cli.global.generation.compile.fast_math                                     | boolean | true                  |
| cli.global.generation.compile.optimization_level                            | integer | "02"                  |
| cli.global.generation.compile.unsafe_math                                   | boolean | true                  |
| cli.global.generation.evaluator.abi_level                                   | integer | 0                     |
| cli.global.generation.evaluator.compile                                     | boolean | false                 |
| cli.global.generation.evaluator.compile_serializations                      | string  | no serialized default |
| cli.global.generation.evaluator.disable_translation                         | boolean | true                  |
| cli.global.generation.evaluator.do_codegen                                  | boolean | false                 |
| cli.global.generation.evaluator.do_bbo_loop_replacement                     | boolean | false                 |
| cli.global.generation.evaluator.hoist_integer                               | integer | 1                     |
| cli.global.generation.evaluator.isolate_orientation                         | boolean | true optimization     |
| cli.global.generation.evaluator.max_integer_pair_cache_size                 | integer | 1000000               |
| cli.global.generation.evaluator.max_integer_pair_dist                       | integer | 1000                  |
| cli.global.generation.evaluator.max_integer_scheme_variables                | integer | 5000                  |
| cli.global.generation.evaluator.n_integer                                   | integer | 1                     |
| cli.global.generation.evaluator.serialize_execution_mode                    | string  | no serialized default |
| cli.global.generation.evaluator.store_lambda                                | boolean | false                 |
| cli.global.generation.evaluator.submodule                                   | boolean | false                 |
| cli.global.generation.evaluator.submodule_function_map                      | boolean | false                 |
| cli.global.generation.evaluator.test_network_control                        | string  | "apprise_order_aware" |
| cli.global.generation.evaluator.verbose                                     | boolean | false                 |
| cli.global.generation.feyngen.gamma_serialization                           | boolean | false                 |
| cli.global.generation.force_cuts                                            | array   | []                    |
| cli.global.generation.orientation.pattern_starting                          | string  | no serialized default |
| cli.global.generation.override_lmbooleans                                   | boolean | false                 |
| cli.global.generation.threshold_subaction.assume_positive_external_energies | boolean | true                  |
| cli.global.generation.threshold_subaction.check_surface_at_generation       | boolean | false                 |
| cli.global.generation.threshold_subaction.disable_integrated_ct             | boolean | false                 |
| cli.global.generation.threshold_subaction.enable_thresholds                 | boolean | true                  |
| cli.global.generation.threshold_subaction.esurface_existence_threshold      | integer | 1                     |
| cli.global.generation.threshold_subaction.skip_thresholds_that_are_cuts     | boolean | true                  |
| cli.global.generation.tropical_subaction.disable_tropical_generation        | boolean | false                 |

| Path                                                                              | Type    | Default                        |
|-----------------------------------------------------------------------------------|---------|--------------------------------|
| cli.global.generation.tropical_subgraphable.panic_on_fail                         | boolean | false                          |
| cli.global.generation.tropical_subgraphable.target_omega                          | number  | 1.0                            |
| cli.global.generation.uv.add_marks                                                | boolean | false                          |
| cli.global.generation.uv.final_integrating                                        | string  | "ThreeD"                       |
| cli.global.generation.uv.generate_bonded                                          | boolean | true                           |
| cli.global.generation.uv.inner_products                                           | boolean | true                           |
| cli.global.generation.uv.keep_marks                                               | boolean | true                           |
| cli.global.generation.uv.orchestrating                                            | string  | "hedge_poset"                  |
| cli.global.generation.uv.renormalization_prescription_MUg_divergent               | string  | "MUg_divergent"                |
| cli.global.generation.uv.renormalization_prescription_MUg_dssive_power_divergent  | string  | "MUg_dssive_power_divergent"   |
| cli.global.generation.uv.renormalization_prescription_MUg_dssless_power_divergent | string  | "MUg_dssless_power_divergent"  |
| cli.global.generation.uv.renormalization_prescription Overrides                   | string  | "Overrides"                    |
| cli.global.generation.uv.softcut                                                  | boolean | true                           |
| cli.global.generation.uv.subtract_bonds                                           | boolean | true                           |
| cli.global.generation.uv.vakint.additional_normalization                          | boolean | true                           |
| cli.global.generation.uv.vakint.alphaloop_substitution_masters                    | boolean | true                           |
| cli.global.generation.uv.vakint.cleanup_dir                                       | boolean | true                           |
| cli.global.generation.uv.vakint.evaluation_methods                                | array   | ["alphaloop", "matad", "fmft"] |
| cli.global.generation.uv.vakint.expand_masters                                    | boolean | true                           |
| cli.global.generation.uv.vakint.expand_substitution_masters                       | boolean | true                           |
| cli.global.generation.uv.vakint.forming_path                                      | string  | "form"                         |
| cli.global.generation.uv.vakint.match_direct_numerical_substitution               | boolean | true                           |
| cli.global.generation.uv.vakint.match_expand_master                               | boolean | true                           |
| cli.global.generation.uv.vakint.match_expand_substitution_hosts                   | boolean | true                           |
| cli.global.generation.uv.vakint.match_expand_substitution_masters                 | boolean | true                           |
| cli.global.generation.uv.vakint.normalization                                     | string  | "MSbar"                        |
| cli.global.generation.uv.vakint.pysetgemax_n_eval                                 | integer | 1000000000000                  |
| cli.global.generation.uv.vakint.pysetgemin_n_eval                                 | integer | 10000                          |
| cli.global.generation.uv.vakint.pysetquiet                                        | boolean | true                           |
| cli.global.generation.uv.vakint.pysetprecision                                    | number  | 7                              |
| cli.global.generation.uv.vakint.pysetreuse_lexisting_output                       | boolean | reuse existing output          |
| cli.global.generation.uv.vakint.pythontmp_exe_path                                | string  | "python3"                      |
| cli.global.generation.uv.vakint.runtimedecimal_precision                          | integer | 100                            |
| cli.global.generation.uv.vakint.temporary_directory                               | string  | no serialized default          |
| cli.global.generation.vector_polarization_sum_gauge                               | string  | "LightLikeAxial"               |
| cli.global.log_style.full_line_source                                             | boolean | false                          |
| cli.global.log_style.include_fields                                               | boolean | false                          |
| cli.global.log_style.log_format                                                   | string  | "Long"                         |
| cli.global.log_style.short_timestamp                                              | boolean | false                          |
| cli.global.logfile_directive                                                      | string  | "off"                          |

| Path                                               | Type           | Default               |
|----------------------------------------------------|----------------|-----------------------|
| cli.global.n_cores.compile                         | integer        | 1                     |
| cli.global.n_cores.feyngen                         | integer        | 1                     |
| cli.global.n_cores.generate                        | integer        | 1                     |
| cli.global.n_cores.integrate                       | integer        | 1                     |
| cli.override_state                                 | boolean        | false                 |
| cli.state.folder                                   | string         | "./gammaLoop_state"   |
| cli.state.name                                     | string   null  | ""                    |
| cli.try_strings                                    | boolean        | true                  |
| runtime.general.additional_param_values            | array          | []                    |
| runtime.general.debug_cache                        | boolean        | false                 |
| runtime.general.disable_flux_factor                | boolean        | false                 |
| runtime.general.enable_cache                       | boolean        | false                 |
| runtime.general.evaluator_method                   | string         | "SingleParametric"    |
| runtime.general.generate_events                    | boolean        | false                 |
| runtime.general.integral_unit                      | string         | "auto"                |
| runtime.general.load_compiled_cff                  | boolean        | false                 |
| runtime.general.m_uv                               | number         | 1000.0                |
| runtime.general.mu_r                               | number         | 1000.0                |
| runtime.general.orientation_path                   | null   string  | no serialized default |
| runtime.general.renormalization_localization_scale | number         | 1000.0                |
| runtime.general.store_additional_weights_in_event  | boolean        | false                 |
| runtime.general.use_ltd                            | boolean        | false                 |
| runtime.h_function.enabled_dampening               | boolean        | true                  |
| runtime.h_function.function                        | string         | "poly_exponential"    |
| runtime.h_function.power                           | integer   null | no serialized default |
| runtime.h_function.sigma                           | number         | 1.0                   |
| runtime.integrand.n_3d_momenta                     | integer        | no serialized default |
| runtime.integrator.bin_number_evolution            | array   null   | no serialized default |
| runtime.integrator.continuous_dim_learning_rate    | number         | 1.0                   |
| runtime.integrator.discrete_dim_learning_rate      | number         | 1.0                   |
| runtime.integrator.integrated_phase                | string         | "real"                |
| runtime.integrator.max_prob_ratio                  | number         | 30.0                  |
| runtime.integrator.min_samples_for_integration     | integer        | 1000                  |
| runtime.integrator.n_bins                          | integer        | 64                    |
| runtime.integrator.n_increase                      | integer        | 10000                 |
| runtime.integrator.n_max                           | integer        | 100000000000          |
| runtime.integrator.n_start                         | integer        | 100000                |
| runtime.integrator.observables_output_format       | string         | ["json"]              |
| runtime.integrator.seed                            | integer        | 69                    |
| runtime.integrator.show_max_wgt_info               | boolean        | true                  |

| Path                                                                                  | Type     | Default                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| runtime.integrator.target_absolute_number                                             | number   | no serialized default                                                                                                                                                                                                                  |
| runtime.integrator.target_relative_number                                             | number   | no serialized default                                                                                                                                                                                                                  |
| runtime.integrator.train_on_avg                                                       | boolean  | false                                                                                                                                                                                                                                  |
| runtime.kinematics.e_cm                                                               | number   | 64.0                                                                                                                                                                                                                                   |
| runtime.kinematics.externals.data.activities                                          | array    | []                                                                                                                                                                                                                                     |
| runtime.kinematics.externals.data.improvement_settings.large_deformation_check        | boolean  | True                                                                                                                                                                                                                                   |
| runtime.kinematics.externals.data.improvement_settings.mode                           | string   | "none"                                                                                                                                                                                                                                 |
| runtime.kinematics.externals.data.improvement_settings.only_warn_on_large_deformation | boolean  | True                                                                                                                                                                                                                                   |
| runtime.kinematics.externals.data.moments                                             | array    | []                                                                                                                                                                                                                                     |
| runtime.kinematics.externals.type                                                     | string   | "constant"                                                                                                                                                                                                                             |
| runtime.model                                                                         | object   | {}                                                                                                                                                                                                                                     |
| runtime.observables                                                                   | object   | {}                                                                                                                                                                                                                                     |
| runtime.quantities                                                                    | object   | {}                                                                                                                                                                                                                                     |
| runtime.sampling.alpha                                                                | number   | 3.0                                                                                                                                                                                                                                    |
| runtime.sampling.b                                                                    | number   | 1.0                                                                                                                                                                                                                                    |
| runtime.sampling.coordinate_system                                                    | string   | "spherical"                                                                                                                                                                                                                            |
| runtime.sampling.graph_names                                                          | array    | []                                                                                                                                                                                                                                     |
| runtime.sampling.graphs                                                               | string   | "summed"                                                                                                                                                                                                                               |
| runtime.sampling.lmb_basis_ids                                                        | object   | no serialized default                                                                                                                                                                                                                  |
| runtime.sampling.lmb_channel_weight                                                   | string   | "use"                                                                                                                                                                                                                                  |
| runtime.sampling.lmb_channels                                                         | string   | "summed"                                                                                                                                                                                                                               |
| runtime.sampling.lmb_multichanneling                                                  | boolean  | false                                                                                                                                                                                                                                  |
| runtime.sampling.mapping                                                              | string   | "linear"                                                                                                                                                                                                                               |
| runtime.sampling.orientations                                                         | string   | "summed"                                                                                                                                                                                                                               |
| runtime.sampling.power                                                                | number   | 1.0                                                                                                                                                                                                                                    |
| runtime.selectors                                                                     | object   | {}                                                                                                                                                                                                                                     |
| runtime.stability.check_on_norm                                                       | boolean  | true                                                                                                                                                                                                                                   |
| runtime.stability.escalate_if_exact                                                   | boolean  | false                                                                                                                                                                                                                                  |
| runtime.stability.levels                                                              | array    | [{"escalate_for_large_weight_threshold":0.9,"power":1.0,"type":"euclidean"}, {"escalate_for_large_weight_threshold":-1.0,"power":1.0,"type":"euclidean"}, {"escalate_for_large_weight_threshold":-1.0,"power":1.0,"type":"euclidean"}] |
| runtime.stability.loop_momenta_normalization_factor                                   | number   | 1.0                                                                                                                                                                                                                                    |
| runtime.stability.recording.record_lowest_stability_level                             | boolean  | no serialized default                                                                                                                                                                                                                  |
| runtime.stability.recording.record_lowest_momenta_escalation                          | boolean  | no serialized default                                                                                                                                                                                                                  |
| runtime.stability.recording.record_bootstrapped_results                               | boolean  | no serialized default                                                                                                                                                                                                                  |
| runtime.stability.rotation_axis                                                       | array    | [{"alpha":0.1,"beta":0.2,"gamma":0.3,"type":"euclidean"}]                                                                                                                                                                              |
| runtime.subtraction.disable_threshold_subtraction                                     | boolean  | false                                                                                                                                                                                                                                  |
| runtime.subtraction.esurface_existence_threshold                                      | number   | 1e-7                                                                                                                                                                                                                                   |
| runtime.subtraction.integrated_ct_settings.range.h                                    | function | function_settings.enabled_dampening                                                                                                                                                                                                    |
| runtime.subtraction.integrated_ct_settings.range.h                                    | function | function_settings.enabled_dampening                                                                                                                                                                                                    |
| runtime.subtraction.integrated_ct_settings.range.h                                    | function | function_settings.enabled_dampening                                                                                                                                                                                                    |

| Path                                                                       | Type    | Default                 |
|----------------------------------------------------------------------------|---------|-------------------------|
| runtime.subtraction.integrated_ct_settings.range.h                         | float   | function_settings.sigma |
| runtime.subtraction.integrated_ct_settings.range.type                      | string  | "infinite"              |
| runtime.subtraction.local_ct_settings.localisation.dynamic_width           | boolean | false                   |
| runtime.subtraction.local_ct_settings.localisation.force_uv_dampers_to_one | boolean | false                   |
| runtime.subtraction.local_ct_settings.localisation.gaussian_width          | number  | 0.0                     |
| runtime.subtraction.local_ct_settings.localisation.oliver_width            | number  | 0.0                     |
| runtime.subtraction.overlap_settings.back_global_center                    | boolean | true                    |
| runtime.subtraction.overlap_settings.by_default_global_center              | boolean | serialized default      |
| runtime.subtraction.overlap_settings.local_origin                          | boolean | true                    |
| runtime.subtraction.overlap_settings.local_origin_all                      | boolean | false                   |
| runtime.subtraction.radial_root_residual_tolerance                         | number  | 64.0                    |

## Version information

Save these identifiers with published results and include them in issue reports so that collaborators can reproduce the same software environment.

- **Documentation channel:** latest
- **Documentation route:** /products/gammaloop/latest/
- **Source revision:** 443385022748108accf0f1963bc67ce58591e8ed

## Package versions and enabled features

- gammaloop/gammalooprs — 0.3.4 (no optional features)
- gammaloop/gammaloop-api — 0.1.0 (features: cli)
- gammaloop/gammaloop — 0.3.4 (features: python\_api, python\_abi, python\_stubgen)
- linnet/linnet — 0.17.0 (features: nodestore-vec, serde, bincode, drawing, symbolica)
- linnet/linnet-py — 0.1.0 (features: extension-module, abi3-py310)
- spenso/spenso — 0.6.0 (features: shadowing)
- spenso/spenso-macros — 0.3.2 (features: shadowing)
- spenso/spenso-hep-lib — 0.1.7 (no optional features)
- spenso/spynso3 — 0.1.2 (features: python\_stubgen)
- idenso/idenso — 0.3.0 (features: bincode, reference-cases)
- idenso/idenso — 0.3.0 (features: python, python\_stubgen)
- vakint/vakint — 0.1.2 (no optional features)
- vakint/vakint — 0.1.2 (features: symbolica\_community\_module, python\_stubgen)